

Quantum Bilisim Ders 6

EPR Paradoxu

Einstein Podolsky Rosen

Solvay Konferansı 1927

Rosen
Bartlett
PBR
Russek

Einstein

Schrodinger
de Broglie

Bohr

Heisenberg

Pauli
Dirac

Kuantum Mekaniğin Belirsizliği olmaksızın

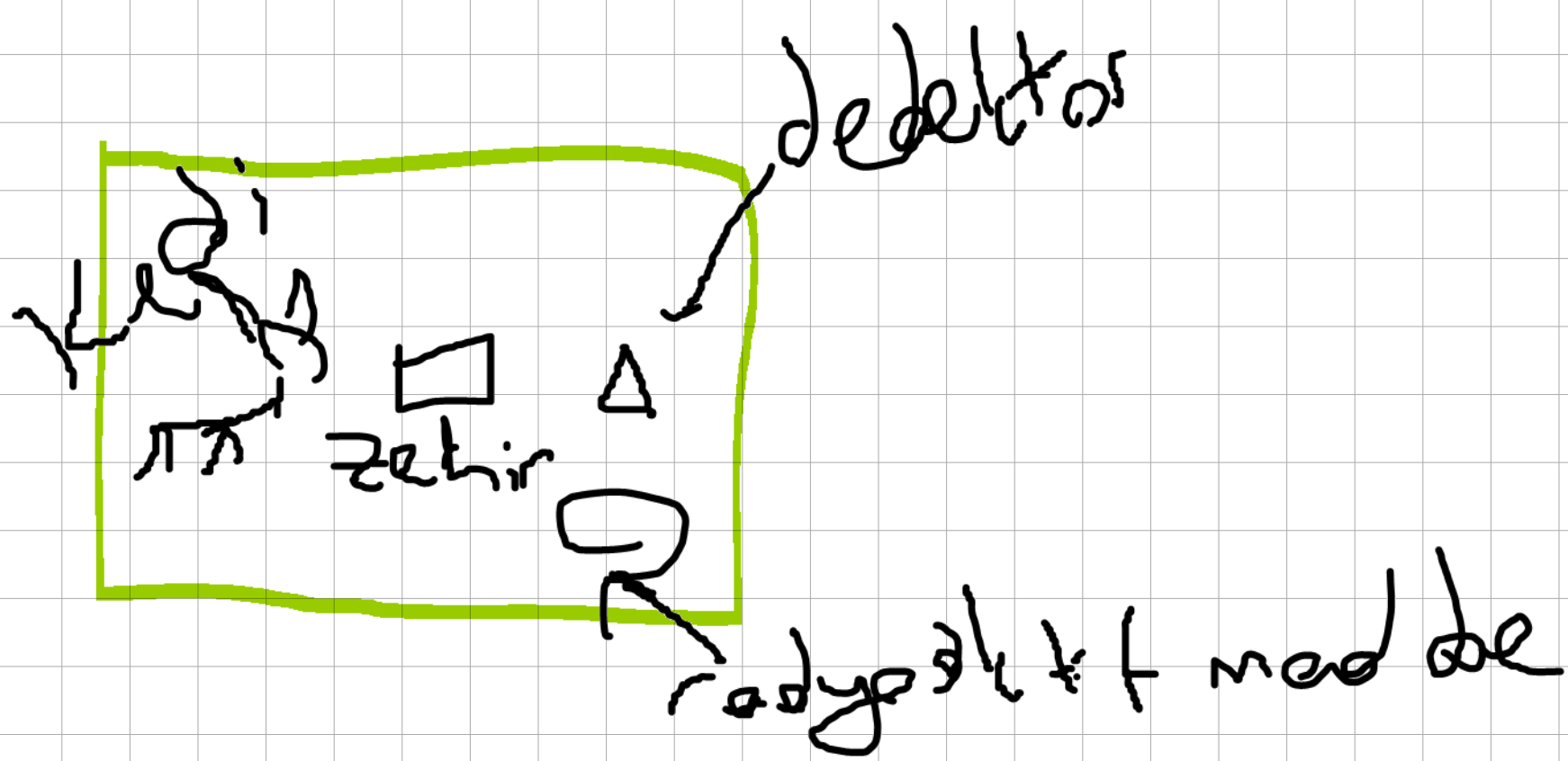
2 tür olasılık → Klasik Olasılık, para

elektron

→ Kuantum Olasılık

Schrodinger Kedi

Von Neumann

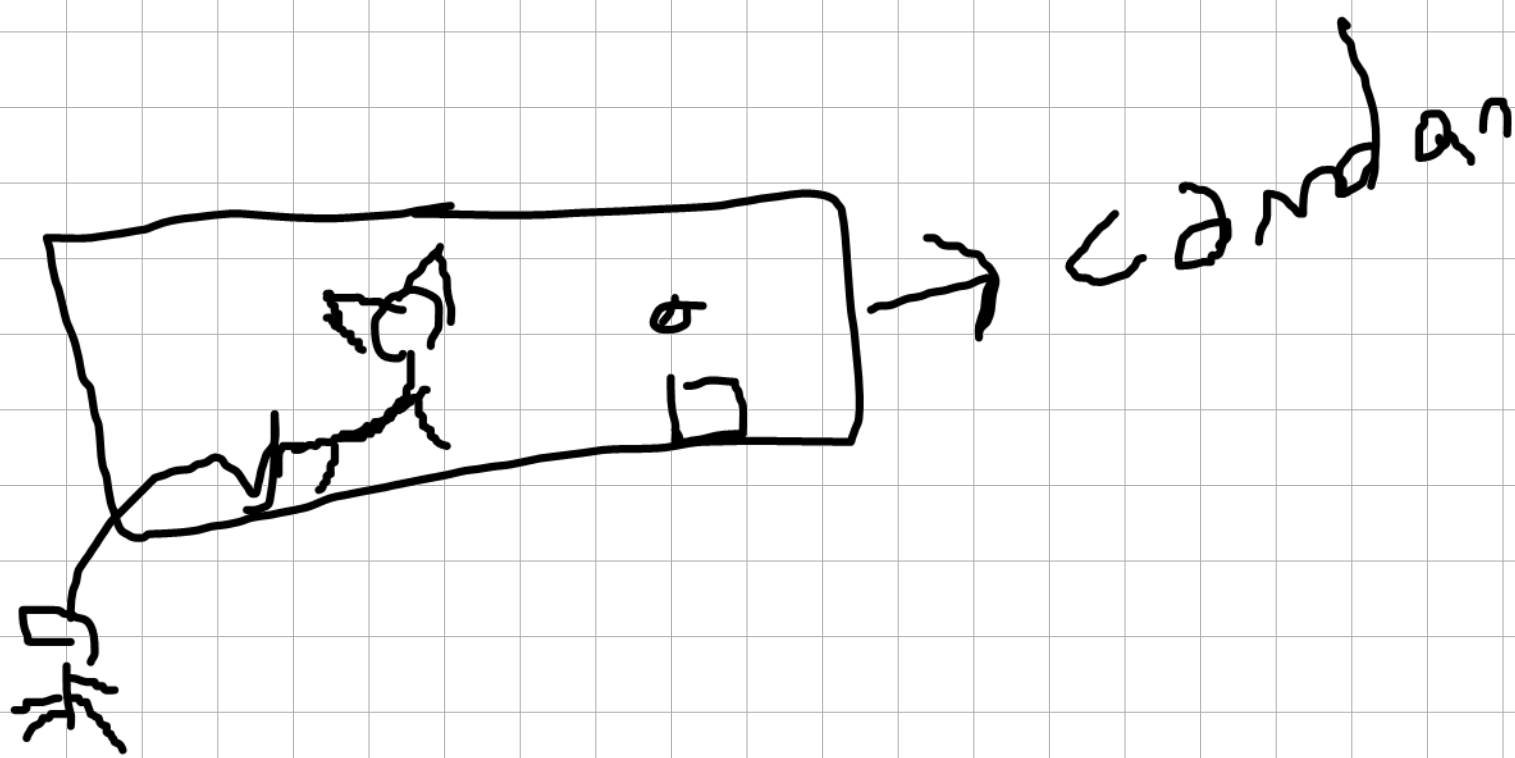


kedi bir zamanda
ölüp de yaşıyor

$$|kedi\rangle = \frac{1}{\sqrt{2}} (|ölü\rangle + |canlı\rangle)$$

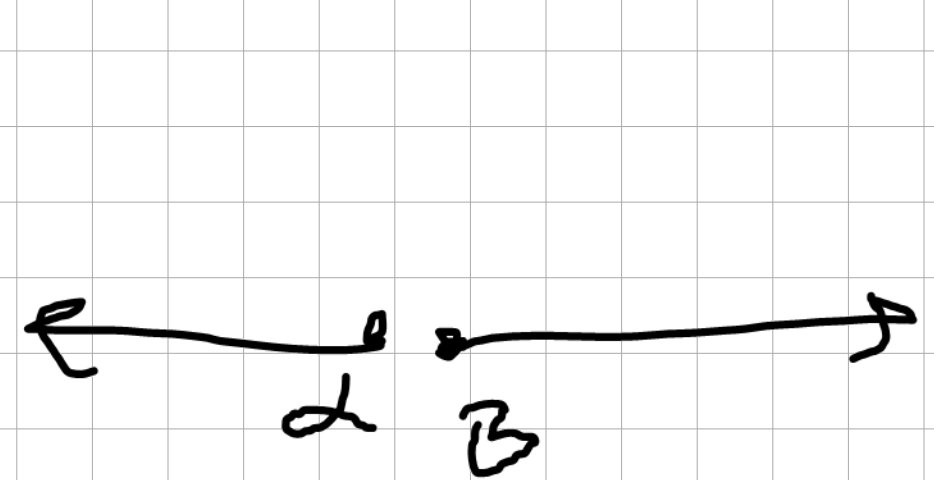
$$|ölü\rangle$$

Wigner



EPR Paradoxu

1935 2 ve 3 2 sistem parçacık
t=0 da etkileşim var



$$\hat{Q}_A \quad \hat{P}_A$$

$$\hat{Q}_B \quad \hat{P}_B$$

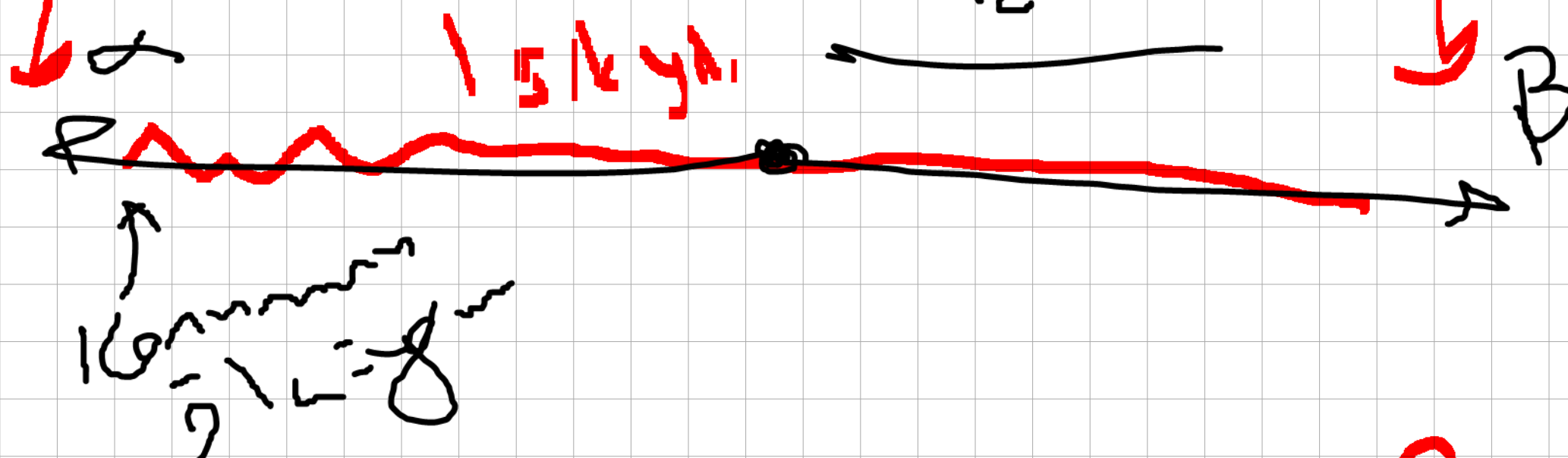
$$[Q_A, P_A] \neq 0$$

$$[Q_B, P_B] \neq 0$$

Heisenberg Belirsizliği
Einstein diyebilmek

$$[Q_A - Q_B, P_A + P_B] = [Q_A, P_A] + [Q_A, P_B] - [Q_B, P_A] - [Q_B, P_B]$$

$$P_A = -P_B$$



David Bohm EPR Varyasyonu

$$|0,0\rangle = \frac{1}{\sqrt{2}} (|\frac{1}{2}, -\frac{1}{2}\rangle - |-\frac{1}{2}, +\frac{1}{2}\rangle)$$

X elemanı

A

A

A

A

Gök uzak

↓

↓

↓

↓

↓

↓

X Z

[S_x, S_y] = iħ S_z

ΔS_x ΔS_y ≠ 0

